

Numerical Analysis Homework 2

Yilin Chen 3230101092 *

(Numerical Analysis, Math2301), Zhejiang University

Due time:2025/10/27 October 21, 2025

I. Linear Interpolation Error

I-a

According to Lagrange formula, $p_1(f; x) = f(x_0) \frac{x - x_1}{x_0 - x_1} + f(x_1) \frac{x - x_0}{x_1 - x_0}$, so

$$f''(\xi(x)) = \frac{2}{\xi^3(x)} = 2 \frac{f(x) - p_1(f; x)}{(x - x_0)(x - x_1)} = 2 \frac{\frac{1}{x} + \frac{x}{2} - \frac{3}{2}}{(x - 1)(x - 2)} = \frac{1}{x}$$

Thus $\xi(x) = \sqrt[3]{2x}$

I-b

Because $\xi(x) = \sqrt[3]{2x}$ and $f''(\xi(x)) = \frac{1}{x}$, we know the 2 function are increasing and decreasing on $[1, 2]$ respectively, thus

$$\max \xi(x) = \xi(2) = \sqrt[3]{4}$$

$$\min \xi(x) = \xi(1) = \sqrt[3]{2}$$

$$\max f''(\xi(x)) = \frac{1}{1} = 1$$

II. Non-negative polynomial interpolation

Imitate the construction of Lagrange formula, let

$$p = \sum_{i=1}^n f_i \left(\frac{\prod_{j \neq i, j=0}^n (x - x_j)}{\prod_{j \neq i, j=0}^n (x_i - x_j)} \right)^2$$

Obviously, p satisfies the requirement.

III. Divided differences of exponential function

III-a

Base case: For $n = 0$,

$$f[t] = e^t = \frac{(e - 1)^0}{0!} e^t.$$

*Electronic address: yilinch@zju.edu.cn

For $n = 1$,

$$f[t, t+1] = \frac{e^{t+1} - e^t}{(t+1) - t} = e^t(e-1) = \frac{(e-1)^1}{1!} e^t.$$

Inductive step: Assume the formula holds for n , then

$$f[t+1, \dots, t+n+1] = \frac{(e-1)^n}{n!} e^{t+1}, \quad f[t, \dots, t+n] = \frac{(e-1)^n}{n!} e^t.$$

From theorem 2.17:

$$f[t, \dots, t+n+1] = \frac{f[t+1, \dots, t+n+1] - f[t, \dots, t+n]}{(t+n+1) - t}.$$

Thus,

$$f[t, \dots, t+n+1] = \frac{(e-1)^n}{n!} \cdot \frac{e^{t+1} - e^t}{n+1} = \frac{(e-1)^n}{n!} \cdot \frac{e^t(e-1)}{n+1} = \frac{(e-1)^{n+1}}{(n+1)!} e^t.$$

This completes the induction.

III-b

From Corollary 2.22, there exists $\xi \in (0, n)$ such that

$$f[0, 1, \dots, n] = \frac{1}{n!} f^{(n)}(\xi).$$

Since $f^{(n)}(x) = e^x$, we have

$$f[0, 1, \dots, n] = \frac{1}{n!} e^\xi.$$

From the induction result with $t = 0$,

$$f[0, 1, \dots, n] = \frac{(e-1)^n}{n!}.$$

Equating the two expressions:

$$\frac{(e-1)^n}{n!} = \frac{1}{n!} e^\xi \quad \Rightarrow \quad e^\xi = (e-1)^n \quad \Rightarrow \quad \xi = n \ln(e-1).$$

Now, compare $\xi = n \ln(e-1)$ with the midpoint $n/2$. Since $e-1 \approx 1.71828$ and $\ln(e-1) \approx 0.54132$, we have

$$\xi \approx 0.54132n > 0.5n = \frac{n}{2}.$$

Therefore, ξ is located to the right of the midpoint $n/2$.

IV. Interpolation and Minimum Location

IV-a

Given the data points: $f(0) = 5$, $f(1) = 3$, $f(3) = 5$, $f(4) = 12$, we construct the divided difference table:

x	$f[x]$			
0	5			
1	3	-2		
3	5	1	1	
4	12	7	2	0.25

The Newton interpolation polynomial is:

$$p_3(x) = 5 + (-2)(x-0) + 1 \cdot (x-0)(x-1) + 0.25 \cdot (x-0)(x-1)(x-3) = 0.25x^3 - 2.25x + 5$$

IV-b

The data suggest a minimum in $(1, 3)$. We find the critical point of $p_3(x)$:

$$p'_3(x) = 0.75x^2 - 2.25.$$

Setting $p'_3(x) = 0$:

$$0.75x^2 - 2.25 = 0 \Rightarrow x^2 = 3 \Rightarrow x = \sqrt{3}.$$

The second derivative test confirms this is a minimum:

$$p''_3(x) = 1.5x > 0 \text{ for } x > 0.$$

Therefore, the approximate location of the minimum is $x_{\min} = \sqrt{3}$.

V. Divided Difference of $f(x) = x^7$

V-a

We compute the divided difference using the recursive definition and the fact that for repeated nodes, the divided difference involves derivatives.

Given $f(x) = x^7$, we have:

$$f(0) = 0, \quad f(1) = 1, \quad f(2) = 128.$$

The derivatives are:

$$f'(x) = 7x^6, \quad f''(x) = 42x^5.$$

$$\text{From Corollary 2.23, } f[1, 1] = \frac{f'(1)}{1!} = 7, \quad f[2, 2] = \frac{f'(2)}{1!} = 448, \quad f[1, 1, 1] = \frac{f''(1)}{2!} = 21$$

We construct the divided difference table for nodes $0, 1, 1, 1, 2, 2$:

x	$f[\cdot]$				
0	0				
1	1	1			
1	1	7	6		
1	1	7	21	15	
2	128	127	120	99	42
2	128	448	321	201	102

That is, $f[0, 1, 1, 1, 2, 2] = 30$.

V-b

By Corollary 2.22, there exists $\xi \in (0, 2)$ such that

$$f[0, 1, 1, 1, 2, 2] = \frac{f^{(5)}(\xi)}{5!}.$$

We compute $f^{(5)}(x) = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot x^2 = 2520x^2$. Thus,

$$\frac{2520\xi^2}{120} = 21\xi^2 = 30 \Rightarrow \xi^2 = \frac{30}{21} = \frac{10}{7}.$$

Hence, $\xi = \sqrt{\frac{10}{7}}$ (since $\xi > 0$).

VI. Hermite Interpolation and Error Estimation

VI-a

Given the conditions:

$$f(0) = 1, \quad f(1) = 2, \quad f'(1) = -1, \quad f(3) = 0, \quad f'(3) = 0,$$

we construct the Hermite interpolation polynomial $H(x)$ of degree at most 4 using the nodes $0, 1, 1, 3, 3$. The divided difference table is:

x	$f[\cdot]$				
0	1				
1	2	1			
1	2	-1	-2		
3	0	-1	0	$\frac{2}{3}$	
3	0	0	$\frac{1}{2}$	$\frac{1}{4}$	$-\frac{5}{36}$

The polynomial is:

$$H(x) = 1 + 1 \cdot x + (-2) \cdot x(x-1) + \frac{2}{3} \cdot x(x-1)^2 + \left(-\frac{5}{36}\right) \cdot x(x-1)^2(x-3).$$

Evaluating at $x = 2$:

$$H(2) = 1 + 2 + (-2)(2)(1) + \frac{2}{3}(2)(1)^2 + \left(-\frac{5}{36}\right)(2)(1)^2(-1) = \frac{11}{18}$$

VI-b

Given $f \in C^5[0, 3]$ and $|f^{(5)}(x)| \leq M$ on $[0, 3]$, the error formula for Hermite interpolation is:

$$f(x) - H(x) = \frac{f^{(5)}(\xi)}{5!} \cdot (x-0)(x-1)^2(x-3)^2, \quad \xi \in (0, 3).$$

At $x = 2$:

$$(2-0)(2-1)^2(2-3)^2 = 2 \cdot 1 \cdot 1 = 2,$$

so the error is:

$$|f(2) - H(2)| = \left| \frac{f^{(5)}(\xi)}{120} \cdot 2 \right| = \frac{|f^{(5)}(\xi)|}{60} \leq \frac{M}{60}.$$

Thus, the maximum possible error is:

$$\frac{M}{60}.$$

VII. Forward and Backward Differences

VII-a

We prove by induction that

$$\Delta^k f(x) = k!h^k f[x_0, x_1, \dots, x_k],$$

where $x_j = x + jh$.

Base case: For $k = 1$,

$$\Delta f(x) = f(x+h) - f(x) = \frac{f(x_1) - f(x_0)}{x_1 - x_0} \cdot (x_1 - x_0) = f[x_0, x_1] \cdot h = 1!h^1 f[x_0, x_1].$$

Inductive step: Assume the identity holds for some k . Then

$$\Delta^{k+1} f(x) = \Delta^k f(x+h) - \Delta^k f(x).$$

By the inductive hypothesis,

$$\Delta^k f(x+h) = k!h^k f[x_1, x_2, \dots, x_{k+1}], \quad \Delta^k f(x) = k!h^k f[x_0, x_1, \dots, x_k].$$

Thus,

$$\Delta^{k+1} f(x) = k!h^k (f[x_1, \dots, x_{k+1}] - f[x_0, \dots, x_k]).$$

From the property of divided differences,

$$f[x_0, x_1, \dots, x_{k+1}] = \frac{f[x_1, \dots, x_{k+1}] - f[x_0, \dots, x_k]}{x_{k+1} - x_0},$$

and since $x_{k+1} - x_0 = (k+1)h$, we have

$$f[x_1, \dots, x_{k+1}] - f[x_0, \dots, x_k] = (k+1)h f[x_0, x_1, \dots, x_{k+1}].$$

Therefore,

$$\Delta^{k+1} f(x) = k!h^k \cdot (k+1)h f[x_0, x_1, \dots, x_{k+1}] = (k+1)!h^{k+1} f[x_0, x_1, \dots, x_{k+1}].$$

This completes the induction.

VII-b

We prove by induction that

$$\nabla^k f(x) = k!h^k f[x_0, x_{-1}, \dots, x_{-k}],$$

where $x_j = x + jh$ (with negative indices).

Base case: For $k = 1$,

$$\nabla f(x) = f(x) - f(x - h) = \frac{f(x_0) - f(x_{-1})}{x_0 - x_{-1}} \cdot (x_0 - x_{-1}) = f[x_{-1}, x_0] \cdot h.$$

By symmetry of divided differences, $f[x_{-1}, x_0] = f[x_0, x_{-1}]$, so

$$\nabla f(x) = hf[x_0, x_{-1}] = 1!h^1 f[x_0, x_{-1}].$$

Inductive step: Assume the identity holds for some k . Then

$$\nabla^{k+1} f(x) = \nabla^k f(x) - \nabla^k f(x - h).$$

By the inductive hypothesis,

$$\nabla^k f(x) = k!h^k f[x_0, x_{-1}, \dots, x_{-k}], \quad \nabla^k f(x - h) = k!h^k f[x_{-1}, x_{-2}, \dots, x_{-(k+1)}].$$

Thus,

$$\nabla^{k+1} f(x) = k!h^k (f[x_0, x_{-1}, \dots, x_{-k}] - f[x_{-1}, x_{-2}, \dots, x_{-(k+1)}]).$$

From the property of divided differences,

$$f[x_0, x_{-1}, \dots, x_{-(k+1)}] = \frac{f[x_0, x_{-1}, \dots, x_{-k}] - f[x_{-1}, x_{-2}, \dots, x_{-(k+1)}]}{x_0 - x_{-(k+1)}},$$

and since $x_0 - x_{-(k+1)} = (k+1)h$, we have

$$f[x_0, x_{-1}, \dots, x_{-k}] - f[x_{-1}, x_{-2}, \dots, x_{-(k+1)}] = (k+1)hf[x_0, x_{-1}, \dots, x_{-(k+1)}].$$

Therefore,

$$\nabla^{k+1} f(x) = k!h^k \cdot (k+1)hf[x_0, x_{-1}, \dots, x_{-(k+1)}] = (k+1)!h^{k+1} f[x_0, x_{-1}, \dots, x_{-(k+1)}].$$

This completes the induction.

VIII. $\frac{\partial}{\partial x_0} f[x_0, x_1, \dots, x_n]$

We prove by induction that

$$\frac{\partial}{\partial x_0} f[x_0, x_1, \dots, x_n] = f[x_0, x_0, x_1, \dots, x_n]$$

Base case:

$$\frac{\partial}{\partial x_0} f[x_0] = \frac{d}{dx_0} f(x_0) = f'(x_0) = f[x_0, x_0]$$

Inductive step: Assume the identity holds for $n = k$. Then for $n = k+1$,

$$\begin{aligned} \frac{\partial}{\partial x_0} f[x_0, x_1, \dots, x_{k+1}] &= \frac{\partial}{\partial x_0} \frac{f[x_0, \dots, x_{k-1}, x_{k+1}] - f[x_0, \dots, x_{k-1}, x_k]}{x_{k+1} - x_k} \\ &= \frac{f[x_0, x_0, x_1, \dots, x_{k-1}, x_{k+1}] - f[x_0, x_0, x_1, \dots, x_{k-1}, x_k]}{x_{k+1} - x_k} \\ &= f[x_0, x_0, x_1, \dots, x_k, x_{k+1}] \end{aligned}$$

This completes the induction.

IX. Min-Max Problem

We wish to minimize

$$\max_{x \in [a, b]} |a_0 x^n + a_1 x^{n-1} + \cdots + a_n|$$

over all choices of $a_1, a_2, \dots, a_n \in \mathbb{R}$, with $a_0 \neq 0$ fixed.

This is equivalent to finding the monic polynomial of degree n (after scaling) that minimizes the maximum absolute value on the interval $[a, b]$. The solution is given by the Chebyshev polynomial of degree n scaled appropriately.

Let $P(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_n$. Define the linear transformation mapping $[a, b]$ to $[-1, 1]$:

$$x = \frac{b-a}{2}t + \frac{a+b}{2}, \quad t \in [-1, 1].$$

Then

$$P(x) = a_0 \left(\frac{b-a}{2}t + \frac{a+b}{2} \right)^n + \text{lower degree terms.}$$

The leading term becomes $a_0 \left(\frac{b-a}{2} \right)^n t^n$. Define

$$Q(t) = \frac{P(x)}{a_0} = \left(\frac{b-a}{2} \right)^n t^n + \text{lower degree terms.}$$

Let $R(t) = Q(t) / \left(\frac{b-a}{2} \right)^n$. Then $R(t)$ is a monic polynomial of degree n . From Corollary 2.48, it is known that the monic polynomial of degree n that minimizes the maximum absolute value on $[-1, 1]$ is $\frac{1}{2^{n-1}} T_n(t)$, where $T_n(t)$ is the Chebyshev polynomial of degree n , and the minimum value is $\frac{1}{2^{n-1}}$. Thus,

$$\min \max_{t \in [-1, 1]} |R(t)| = \frac{1}{2^{n-1}}.$$

Then

$$\min \max_{t \in [-1, 1]} |Q(t)| = \left(\frac{b-a}{2} \right)^n \cdot \frac{1}{2^{n-1}} = \frac{(b-a)^n}{2^{2n-1}}.$$

Since $\max_{x \in [a, b]} |P(x)| = |a_0| \max_{t \in [-1, 1]} |Q(t)|$, we obtain

$$\min \max_{x \in [a, b]} |P(x)| = \frac{|a_0|(b-a)^n}{2^{2n-1}}.$$

X. Minimax Property of Scaled Chebyshev Polynomial

Let $x'_k = \cos \frac{k}{n} \pi, k = 0, 1, \dots, n$, then T_n have extreme values at the $(n+1)$ points:

$$T_n(x'_k) = (-1)^k$$

Suppose $\|\hat{p}_n\|_\infty \leq \|p\|_\infty$ does not hold. Then Theorem 2.45 implies that

$$\exists p \in \mathbb{P}_n^a, s.t. |p(x)| < \frac{1}{T_n(a)} \quad (1)$$

Let $Q(x) = \frac{T_n(x)}{T_n(a)} - p(x)$, then

$$Q(x'_k) = \frac{(-1)^k}{T_n(a)} - p(x'_k)$$

By (2.46), $Q(x)$ has alternating signs at these $n+1$ points. Hence $Q(x)$ must have n zeros. However, by the construction of $Q(x)$, the degree of $Q(x)$ is at most $n-1$.

Therefore, $Q(x) \equiv 0$ and $p(x) = \frac{T_n(x)}{T_n(a)}$, which implies that

$$\max |p(x)| = \frac{1}{T_n(a)}$$

This is a contradiction to (1).

XI. Proof of Lemma 2.53

According to the definition : $\forall k = 0, 1, \dots, n$, $b_{n,k}(t) := \binom{n}{k} t^k (1-t)^{n-k}$

$$\begin{aligned}
\frac{n-k}{n} b_{n,k}(t) + \frac{k+1}{n} b_{n,k+1}(t) &= \frac{n-k}{n} \frac{n!}{k!(n-k)!} t^k (1-t)^{n-k} + \frac{k+1}{n} \frac{n!}{(k+1)!(n-k-1)!} t^{k+1} (1-t)^{n-k-1} \\
&= \frac{(n-1)!}{k!(n-k-1)!} t^k (1-t)^{n-k} + \frac{(n-1)!}{k!(n-k-1)!} t^{k+1} (1-t)^{n-k-1} \\
&= \frac{(n-1)!}{k!(n-k-1)!} t^k (1-t)^{n-k-1} (1-t+t) \\
&= b_{n-1,k}(t)
\end{aligned}$$

XII. Proof of Lemma 2.55

$$\begin{aligned}
\int_0^1 b_{n,k}(t) dt &= \int_0^1 \binom{n}{k} t^k (1-t)^{n-k} dt \\
&= \frac{n!}{k!(n-k)!} B(k+1, n-k+1) \\
&= \frac{n!}{k!(n-k)!} \frac{\Gamma(k+1)\Gamma(n-k+1)}{\Gamma(n+2)} \\
&= \frac{n!}{k!(n-k)!} \frac{k!(n-k)!}{(n+1)!} \\
&= \frac{1}{n+1}
\end{aligned}$$